Newton School of Technology

Data Visualization & Analytics

Final Project Report

Detailed Academic + Industry Report

10–15 pages — Excluding Appendix

Detailed Academic + Industry Report — 10 to 15 pages, excluding Appendix and Contribution Matrix

1. Cover Page

- Project Title
- Sector
- Team ID, team member names, and roles
- Faculty Mentor name
- Institute: Newton School of Technology
- GitHub Repository URL
- Tableau Public Dashboard URL
- Submission Date

2. Executive Summary (1 page)

- Problem: What business challenge was addressed?
- Approach: How was the data sourced, cleaned, and analysed? (Python + Tableau pipeline)
- Key Insights: 3–4 most important findings
- Key Recommendations: 2–3 highest-impact actions
- Note: Write this for a CEO or policy-maker — no technical jargon

3. Sector & Business Context

- Sector overview and current industry challenges
- Who is the decision-maker this project serves?
- Why was this specific problem chosen?
- What is the business value of solving it?

4. Problem Statement & Objectives

- Formal problem definition
- Project scope — what is in and out of scope
- Success criteria — how will you know the project succeeded?

5. Data Description

- Exact dataset source citation and direct access link
- Data structure: number of rows, columns, and time period
- Column-by-column explanation of key fields
- Data limitations and known biases

6. Data Cleaning & ETL Pipeline

- All primary cleaning and transformation steps were executed in Python (as per Capstone 2 requirement)
- Missing values: how many, in which columns, and how handled
- Outlier detection and treatment method
- Data type conversions, standardisation, and formatting

- Feature engineering — new columns or derived fields created
- Assumptions made during cleaning
- Reference: notebooks/02_cleaning.ipynb committed to GitHub

7. KPI & Metric Framework

- KPI definitions — what each metric measures
- Formula or computation logic (reference the Python notebook)
- Why each KPI matters for the business problem
- Mapping of KPIs to project objectives

8. Exploratory Data Analysis (EDA)

- Trend analysis — how key metrics change over time
- Comparison analysis — performance across segments, regions, or categories
- Distribution analysis — spread and skew of key variables
- Correlation analysis — relationships between variables
- Written insights for each chart — not just descriptions, but what it means for the business
- Reference: notebooks/03_eda.ipynb committed to GitHub

9. Statistical Analysis

- Forecasting (if applicable)
- Segmentation or clustering
- Root cause or regression analysis
- Hypothesis testing (e.g., t-test, chi-squared)
- Risk or anomaly detection
- Scenario analysis
- Reference: notebooks/04_statistical_analysis.ipynb committed to GitHub

10. Tableau Dashboard Design

- Dashboard objective — what decision does it support?
- View structure — executive summary view and operational drill-down view
- Filters and interactive elements included
- Screenshots with explanations of each view
- Tableau Public URL (must be accessible at time of submission)
- Reference: tableau/screenshots/ and tableau/dashboard_links.md in GitHub

11. Insights Summary

- 8–12 most important insights from the full analysis
- Written in decision language — each insight must tell the reader what to think or do
- Not: "Sales dropped in May" — but: "Sales dropped in May due to stockouts in the North region, requiring a change in re-order points"

12. Recommendations

- 3–5 specific, actionable, and prioritised business recommendations
- Each recommendation must map to: Insight → Recommendation → Expected Business Impact
- Feasibility notes — what would it take to implement this?

13. Impact Estimation

- Estimated impact of recommendations on: cost savings, efficiency gains, revenue improvement, or risk reduction
- Approximate numbers are acceptable — the logic must be sound and clearly explained
- Why should the stakeholder act on this now?

14. Limitations

- Data gaps or quality issues that constrain conclusions
- Assumptions made that may not hold in all contexts
- What cannot be concluded from this analysis and why

15. Future Scope

- What additional analysis could be performed with more data or time?
- What new data sources would strengthen the findings?
- Could a predictive model or real-time dashboard be built next?

16. Conclusion

- Final summary of the value delivered by this project
- One paragraph: problem → approach → key insight → recommended action

17. Appendix

- Data dictionary (full column definitions)
- Additional charts or EDA outputs not included in the main report
- Python/SQL logic excerpts (if useful for understanding methodology)
- Raw cleaning log from notebooks/02_cleaning.ipynb

18. Contribution Matrix (Mandatory)

This section must clearly document each team member's contribution across all project phases. Claims must match evidence in GitHub Insights, PR history, and committed files. Any mismatch may result in individual grade adjustments.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Name 1							
Name 2							

Name 3							
Name 4							
Name 5							
Name 6							
Name 7							

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: _____ **Date:** _____